Cambridge (CIE) IGCSE Chemistry
Paper 6: Alternative to Practical (Set A)

Saturday 26 April 2025

Morning (Time: 1 hour 0 minutes)

Total marks

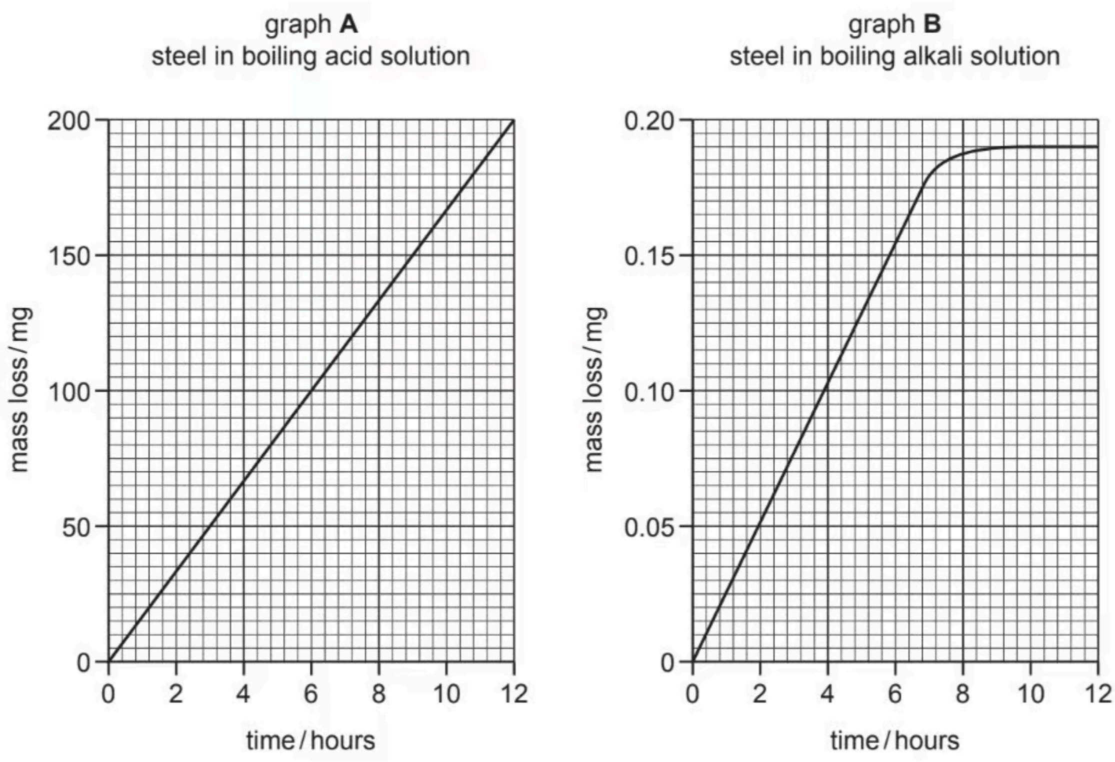
/ 40

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Scan here to mark your mock exam
or visit the mock exams landing page for this course

1 (a) Identical pieces of steel were placed in two different boiling liquids for 12 hours. The graphs show how the mass of each piece of steel changed.



Give one similarity in the change in mass of the steel in both liquids.

....

(1 mark)

(b) Describe two ways in which the mass loss shown in graph **A** is different from that shown in graph **B**.

1. ....

....

2. ....

....

....

....

....

(3 marks)

(c) State **two** different safety precautions that would need to be taken when carrying out this investigation.

1. ....

2. ....

....

....

(2 marks)

2 (a) A student investigated the reaction between aqueous sodium carbonate and two different solutions of dilute hydrochloric acid, **A** and **B**.

The reaction is:

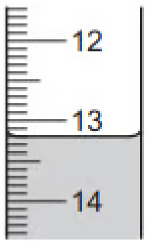
$$Na_2CO_3\,(aq) + 2HCl\,(aq) \rightarrow 2NaCl\,(aq) + H_2O\,(l) + CO_2\,(g)$$

Three experiments were carried out.

Experiment 1

Using a measuring cylinder, 25 cm^3 of aqueous sodium carbonate were poured into a conical flask.

Thymolphthalein indicator was added to the conical flask. A burette was filled up to the 0.0 cm^3 mark with solution **A**. Dilute hydrochloric acid **A** was added to the flask, until the solution just changed colour. Use the burette diagram to record the reading in the table.

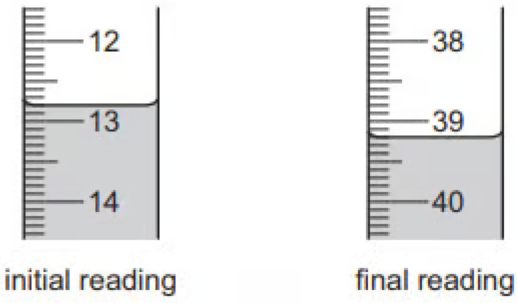


final reading

Experiment 2

Experiment 1 was repeated using methyl orange indicator instead of thymolphthalein.

Methyl orange is red-orange in acidic solutions and yellow in alkaline solutions. Use the burette diagrams to record the readings in the table and complete the table.



	experiment 1	experiment 2
final burette reading / cm^3		
initial burette reading / cm^3		
difference / cm^3		

....

....

....

....

(4 marks)

(b) What colour change was observed in the flask in experiment 2?

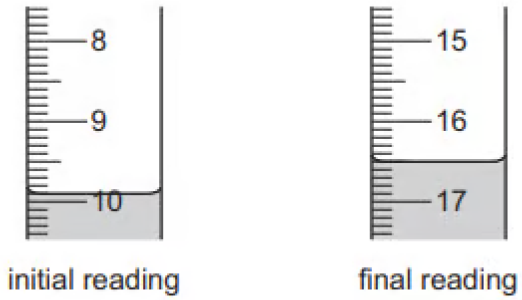
from .... to ....

....

(1 mark)

(c) *Experiment 3*

Experiment 1 was repeated using solution **B** of acid instead of solution **A**. Use the burette diagrams to record the readings in the table and complete the table.



	experiment 3
final burette reading / cm^3	
initial burette reading / cm^3	
difference / cm^3	

....

....

(2 marks)

(d) Suggest **one** observation, other than colour change, that is made when hydrochloric acid is added to sodium carbonate.

....

(1 mark)

(e) Complete the sentence below. Experiment .... needed the largest volume of hydrochloric acid to change the colour of the indicator.

(1 mark)

(f) What would be a more accurate method of measuring the volume of the aqueous sodium carbonate?

(1 mark)

(g) What would be the effect on the results, if any, if the solutions of sodium carbonate were warmed before adding the hydrochloric acid? Give a reason for your answer.

effect on results .... reason ....

(2 marks)

(h) i) Determine the ratio of volumes of dilute hydrochloric acid used in experiments 1 and 3.

[1]

ii) Use your answer to (h)(i) to deduce how the concentration of solution **A** differs from that of solution **B**.

[1]

(2 marks)

(i) Suggest a different method, using standard laboratory chemicals, to determine which of the solutions of dilute hydrochloric acid, **A** or **B**, is more concentrated.

(3 marks)

3 (a) An aqueous solution of substance **X** was analysed. Substance **X** was an iron(III) salt containing one other cation. The tests on **X** and some of the observations are in the following table. Complete the observations in the table

tests	observations
Colour of solution **X**	dark yellow
i) Drops of aqueous sodium hydroxide were added to about 2 cm^3 of the solution. Excess aqueous sodium hydroxide was added to the test-tube. The mixture was heated. The gas given off was tested with damp indicator paper.	.... [3] pungent smell, indicator turned blue,pH 10
ii) Experiment (i) was repeated using aqueous ammonia instead of aqueous sodium hydroxide	.... [2]
To about 2 cm^3 of solution **X** was added dilute sulphuric acid. Two pieces of zinc were added. The mixture was heated and the gas given off tested. After 10 minutes the mixture was filtered and test (a)(i) was repeated	lighted splint popped green precipitate insoluble in excess
A few drops of hydrochloric acid were added to about 2 cm^3 of solution **X**. About 1 cm^3 of barium chloride solution was added to the mixture.	white precipitate

(5 marks)

(b) i) Name the gas given off when sulfuric acid was added to the solution.

[1]

ii) What type of chemical reaction occurs in (d). Explain your answer.

[2]

(3 marks)

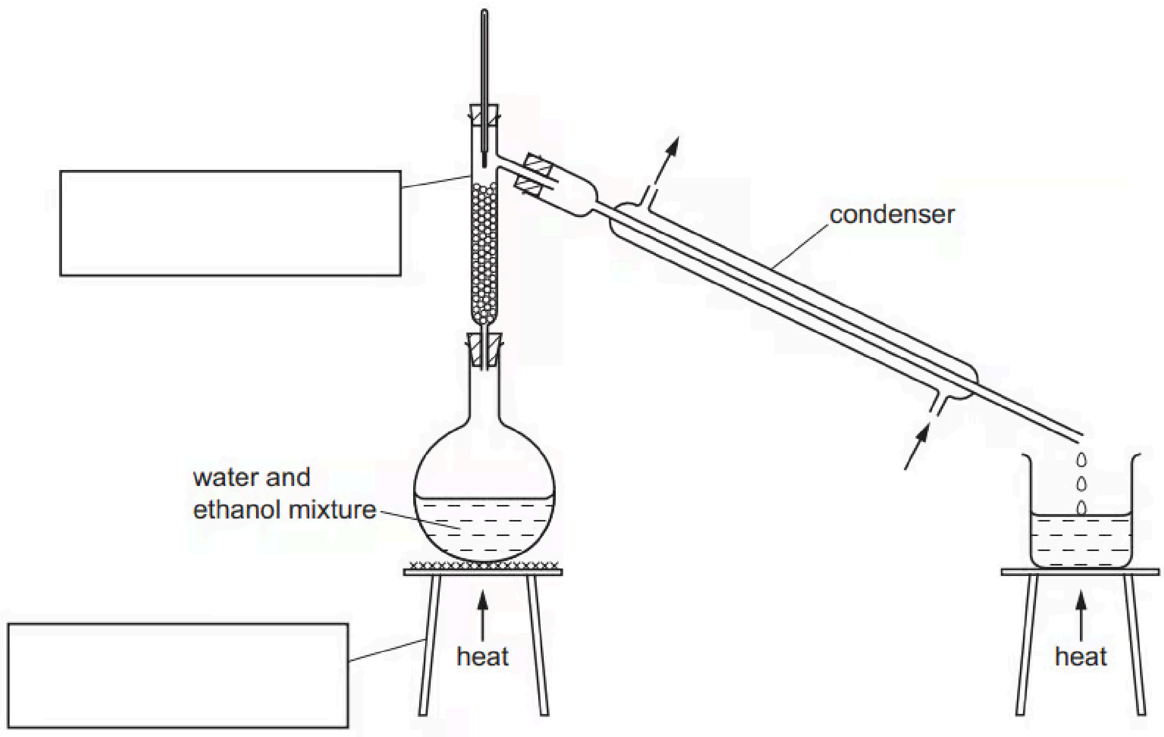
(c) What conclusions can you draw about the anion and the other cation in substance **X**?

anion ....

cation ....

(2 marks)

4 (a) The diagram shows the apparatus used to separate a mixture of water, boiling point 100 °C, and ethanol, boiling point 78 °C.



Complete the boxes to name the apparatus.

....

....

(2 marks)

(b) Label the arrows on the condenser.

....

(1 mark)

(c) Identify **one** mistake in the apparatus.

....

(1 mark)

(d) Which liquid would collect first? Explain your answer.

(2 marks)

(e) Why would it be better to use an electrical heater instead of a Bunsen burner to heat the water and ethanol mixture?

(1 mark)